

Тема: Настройка протоколов динамической маршрутизации RIP v2 и OSPF

1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров. Выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

(nb! Каждое соединение маршрутизатора с маршрутизатором - это отдельная сеть.)

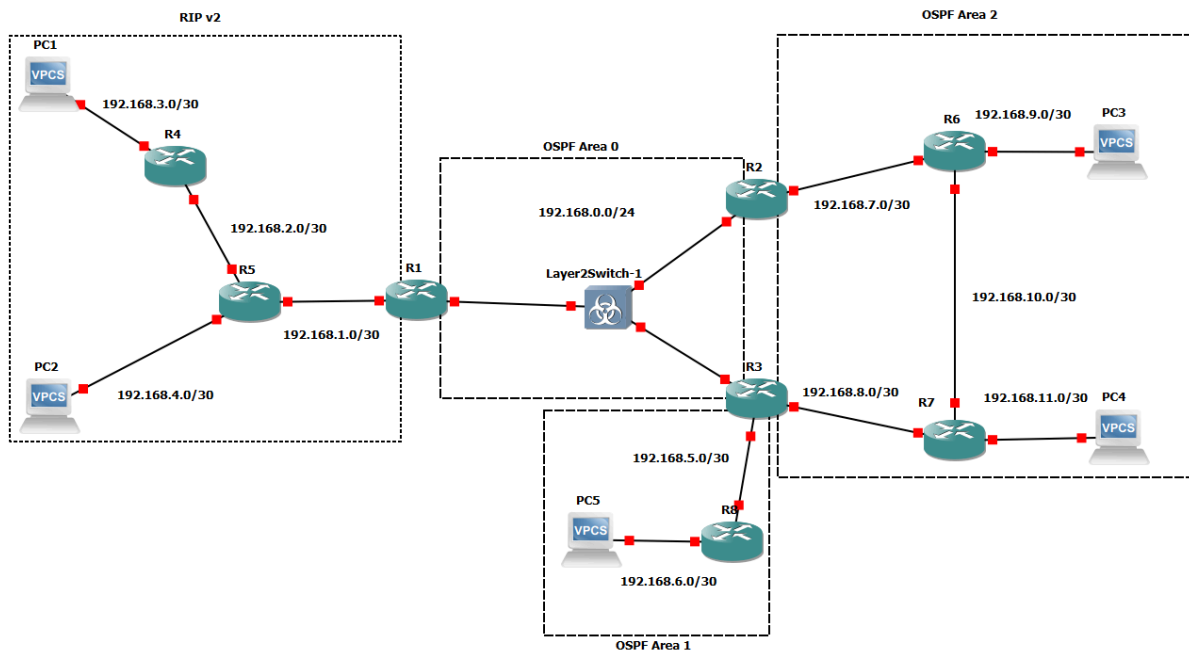


Рисунок 1.1 – Схема сети с подписанными адресами сетей.

В большинстве сетей находятся по два устройства, поэтому для них были выбраны адреса вида 192.168.x.0/30, где x – номер сети.

Назначим статические адреса всем устройствам. Далее приведен пример для PC1 и R4. Остальные конфигурации будут приведены в отдельных файлах.

Для VPCS (PC1):

```
PC1> ip 192.168.3.2 30 192.168.3.1
Checking for duplicate address...
PC1 : 192.168.3.2 255.255.255.252 gateway 192.168.3.1
```

```
PC1> save
Saving startup configuration to startup.vpc
. done
```

Для R4:

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#int f 0/0
R4(config-if)#ip address 192.168.3.1 255.255.255.252
R4(config-if)#no shut
*Mar  1 00:02:40.343: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to
up
R4#
*Mar  1 00:02:41.263: %SYS-5-CONFIG_I: Configured from console by console
*Mar  1 00:02:41.343: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#int f 1/0
R4(config-if)#ip address 192.168.2.2 255.255.255.252
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#
*Mar  1 00:04:42.199: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to
up
*Mar  1 00:04:43.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/0, changed state to up
R4#wr
*Mar  1 00:06:33.359: %SYS-5-CONFIG_I: Configured from console by console
R4#wr
Building configuration...
[OK]
```

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

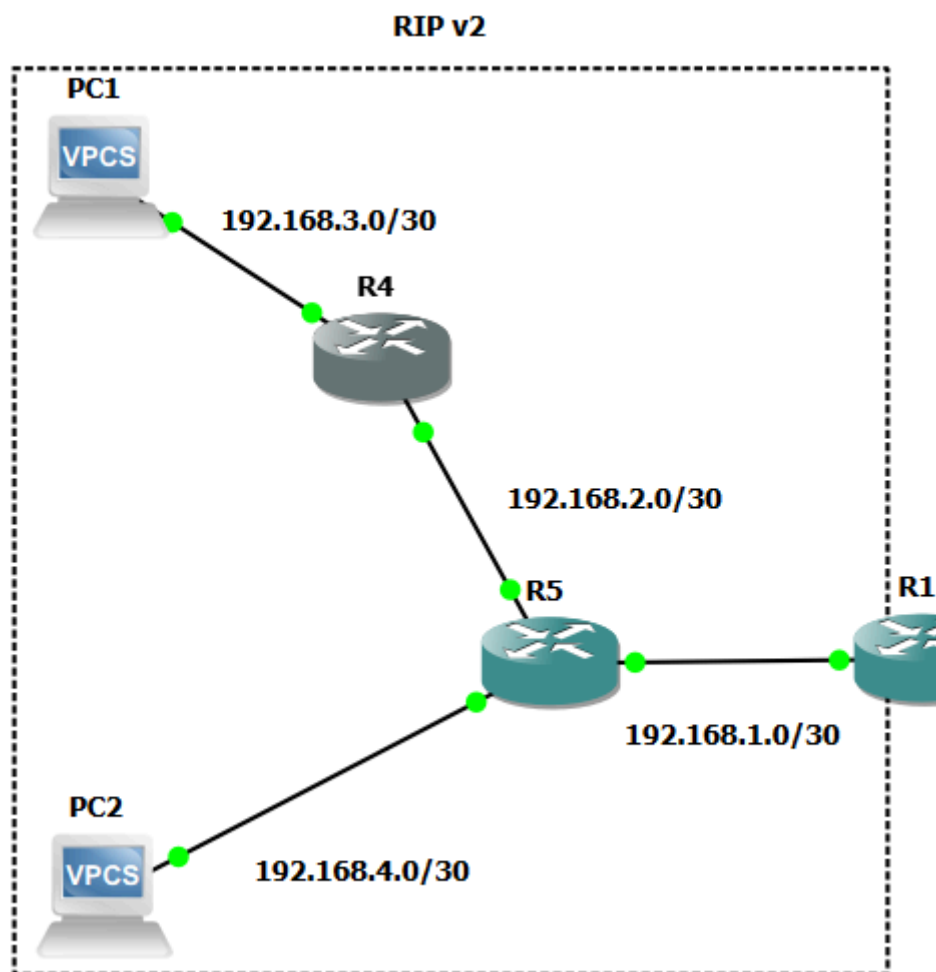


Рисунок 2.1 – Область RIP v2.

Таблица маршрутизации на R4:

```
R4#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    192.168.2.0/30 is subnetted, 1 subnets
C      192.168.2.0 is directly connected, FastEthernet1/0
    192.168.3.0/30 is subnetted, 1 subnets
C      192.168.3.0 is directly connected, FastEthernet0/0
```

Есть подключенные сети, но никаких больше маршрутов не настроено.

```
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.2.0
R4(config-router)#network 192.168.3.0
```

```
R5(config)#router rip
R5(config-router)#version 2
R4(config-router)#no auto-summary
R5(config-router)#network 192.168.2.0
R5(config-router)#network 192.168.4.0
```

После этого проверяем таблицу маршрутизации на R4:

```
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
R       192.168.4.0/30 [120/1] via 192.168.2.1, 00:00:26, FastEthernet1/0
R       192.168.4.0/24 is possibly down,
        routing via 192.168.2.1, FastEthernet1/0
      192.168.2.0/30 is subnetted, 1 subnets
C       192.168.2.0 is directly connected, FastEthernet1/0
      192.168.3.0/30 is subnetted, 1 subnets
C       192.168.3.0 is directly connected, FastEthernet0/0
```

И еще проверяем таблицу на R5:

```
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
      192.168.4.0/30 is subnetted, 1 subnets
C       192.168.4.0 is directly connected, FastEthernet0/0
      192.168.1.0/30 is subnetted, 1 subnets
```

```

C      192.168.1.0 is directly connected, FastEthernet2/0
      192.168.2.0/30 is subnetted, 1 subnets
C      192.168.2.0 is directly connected, FastEthernet1/0
      192.168.3.0/30 is subnetted, 1 subnets
R      192.168.3.0 [120/1] via 192.168.2.2, 00:00:20, FastEthernet1/0

```

```
PC1> ping 192.168.4.2
```

```

84 bytes from 192.168.4.2 icmp_seq=1 ttl=62 time=29.398 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=62 time=25.949 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=62 time=25.187 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=25.717 ms

```

Здесь в каждой таблице теперь есть запись с кодом R – это означает, что теперь R4 знает, хост в сети 192.168.4.0/30 доступен via 192.168.2.1 (R5).

В таблице R4 остался 192.168.4.0/24 от неправильной настройки, потому что маска автоматически поставилась в 24. После перенастройки, через некоторое время к нему добавилась надпись is possibly down, потом он вообще пропал.

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2. Зону 1 настроить как полностью (nb!) тупиковую.

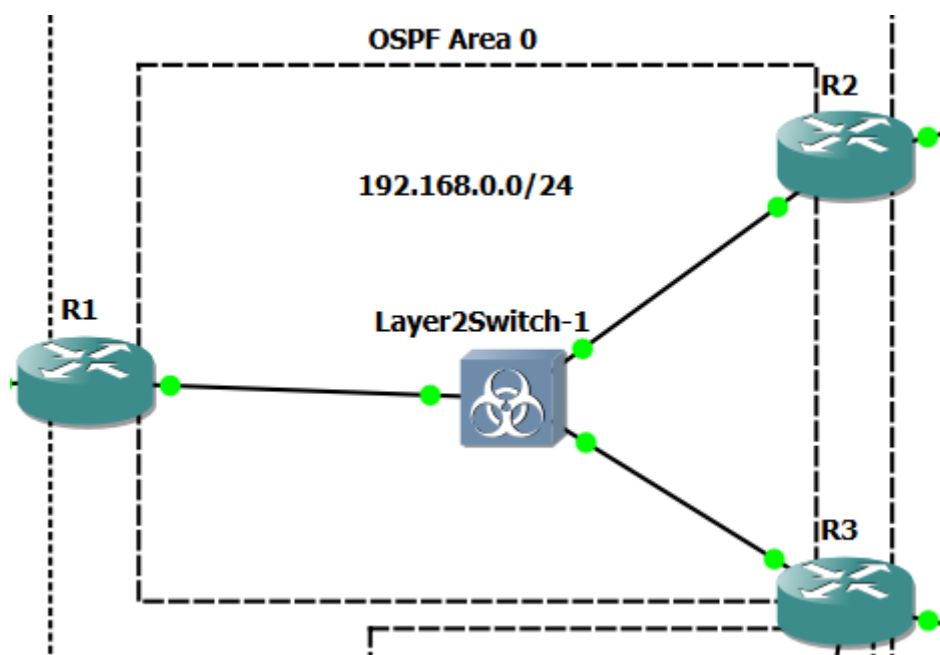


Рисунок 3.1 – OSPF зона 0.

```

R1(config)#router ospf 1
R1(config-router)#router-id 10.0.0.1
R1(config-router)#network 192.168.0.0 ?
A.B.C.D OSPF wild card bits

```

```

R1(config-router)#network 192.168.0.0 0.0.0.255 area 0

R2(config)#router ospf 2
R2(config-router)#router-id 10.0.0.2
R2(config-router)#network 192.168.0.0 0.0.0.255 area 0

R3(config)#router ospf 1
R3(config-router)#router-id 10.0.0.3
R3(config-router)#network 192.168.0.0 0.0.0.255 area 0

```

После настройки пошли Hello сообщения между маршрутизаторами в Area 0. Но таблицы маршрутизации пока пустые.

Захват с Standard input [R1 FastEthernet1/0 to Layer2Switch-1 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
5	4.042991	192.168.0.1	224.0.0.5	OSPF	94	Hello Packet
8	6.248074	192.168.0.2	224.0.0.5	OSPF	94	Hello Packet
16	14.021300	192.168.0.1	224.0.0.5	OSPF	94	Hello Packet
19	16.244610	192.168.0.2	224.0.0.5	OSPF	94	Hello Packet

Рисунок 3.2 – Hello пакеты от R1 и R2.

Настроим зону 1.

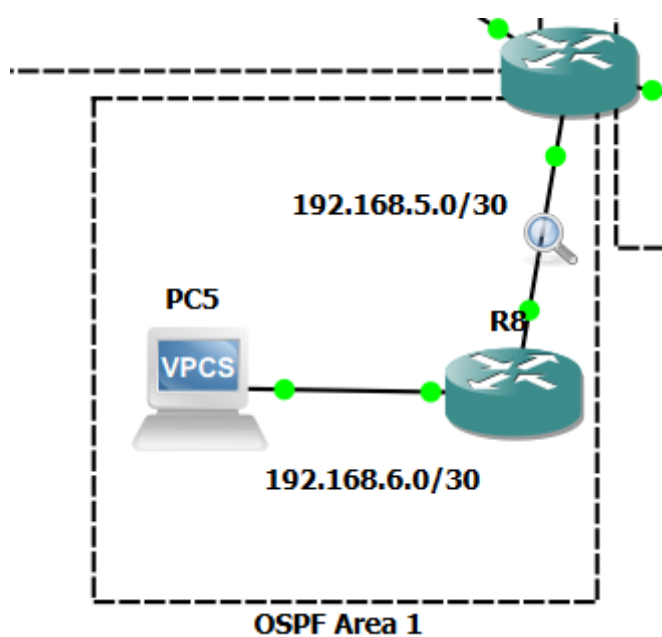


Рисунок 3.3 – OSPF зона 1

```
R3(config)#router ospf 1
R3(config-router)#network 192.168.5.0 0.0.0.255 area 1

R8(config)#router ospf 1
R8(config-router)#router
R8(config-router)#network 192.168.5.0 0.0.0.255 area 1
R8(config-router)#network 192.168.6.0 0.0.0.255 area 1
R8(config-router)#passive-interface fastEthernet 1/0
```

На этом этапе, так как через R3 появилась информация о чужой зоне в таблице R1 появляется маршрут до сети 192.168.5.0/30.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.168.5.0/30 is subnetted, 1 subnets
O IA   192.168.5.0 [110/2] via 192.168.0.3, 00:04:31, FastEthernet1/0
C      192.168.0.0/24 is directly connected, FastEthernet1/0
       192.168.1.0/30 is subnetted, 1 subnets
C      192.168.1.0 is directly connected, FastEthernet0/0
```

Сделаем зону тупиковой, на R3 и R8 пропишем следующее:

```
R3(config-router)#area 1 stub
R8(config-router)#area 1 stub

R8#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.5.1 to network 0.0.0.0

    192.168.5.0/30 is subnetted, 1 subnets
C      192.168.5.0 is directly connected, FastEthernet0/0
       192.168.6.0/30 is subnetted, 1 subnets
C      192.168.6.0 is directly connected, FastEthernet1/0
O IA  192.168.0.0/24 [110/2] via 192.168.5.1, 00:03:31, FastEthernet0/0
O*IA  0.0.0.0/0 [110/2] via 192.168.5.1, 00:03:31, FastEthernet0/0
```

Теперь настроим Area 2.

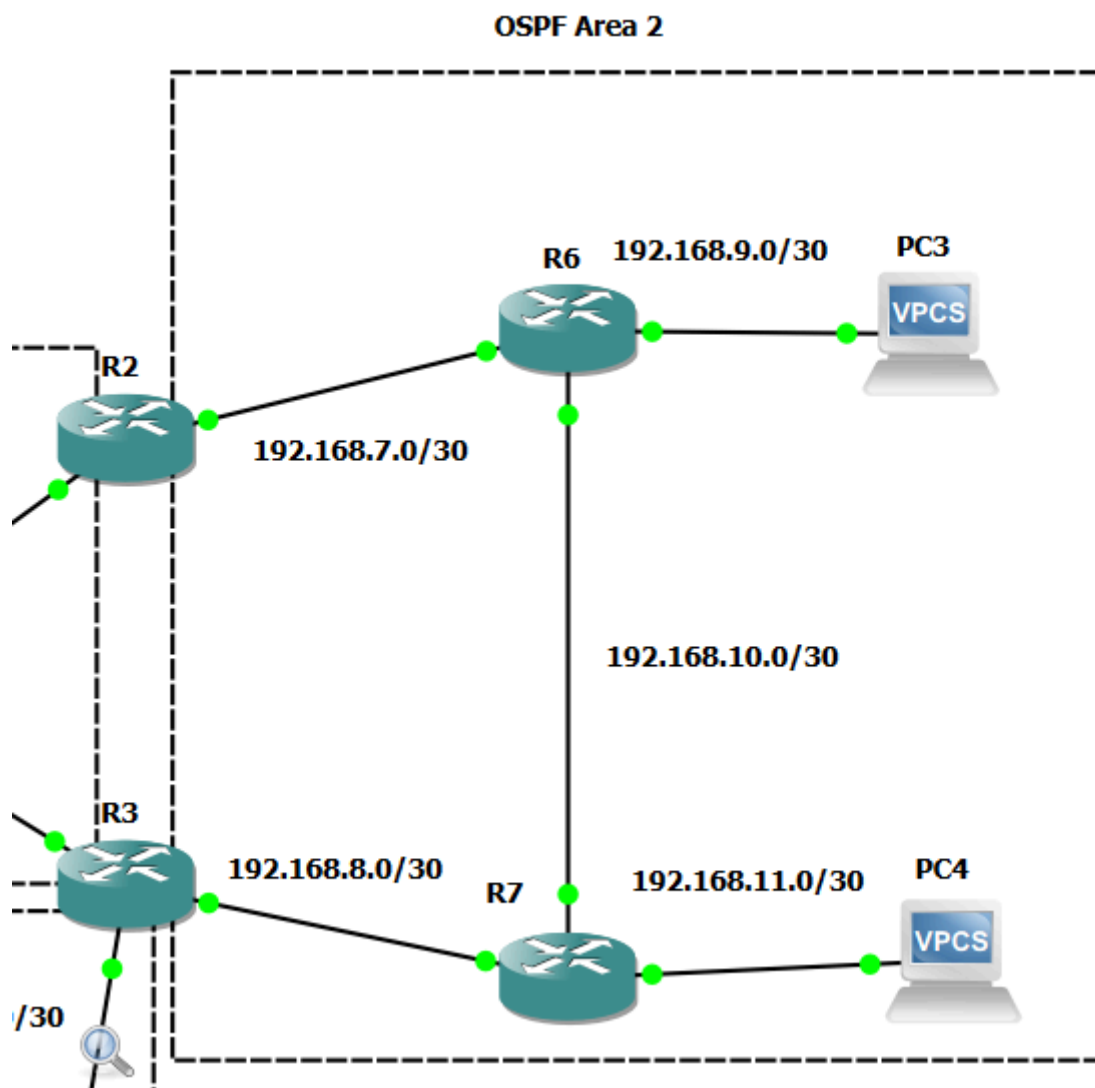


Рисунок 3.4 – OSPF зона 2

```
R2(config)#router ospf 1
R2(config-router)#network 192.168.7.0 0.0.0.255 area 2

R3(config)#router ospf 1
R3(config-router)#network 192.168.8.0 0.0.0.255 area 2

R6(config)#router ospf 1
R6(config-router)#network 192.168.0.0 0.0.255.255 area 2
R6(config-router)#passive-interface fastEthernet 1/0

R7(config)#router ospf 1
R7(config-router)#network 192.168.0.0 0.0.255.255 area 2
R7(config-router)#passive-interface fastEthernet 1/0
```



```
PC3> ping 192.168.11.2
```

```
84 bytes from 192.168.11.2 icmp_seq=1 ttl=62 time=32.409 ms
84 bytes from 192.168.11.2 icmp_seq=2 ttl=62 time=25.279 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=62 time=25.666 ms
84 bytes from 192.168.11.2 icmp_seq=4 ttl=62 time=25.314 ms
```

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network ?
    A.B.C.D  Network number
R1(config-router)#network 192.168.1.0
R1(config-router)#redistribute rip subnets
R1(config)#router rip
R1(config-router)#red
R1(config-router)#redistribute ospf 1 metric 1
```

```
R5(config)#router rip
R5(config-router)#network 192.168.1.0
```

```
R1#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    192.168.8.0/30 is subnetted, 1 subnets
O IA    192.168.8.0 [110/2] via 192.168.0.3, 00:39:26, FastEthernet1/0
    192.168.9.0/30 is subnetted, 1 subnets
O IA    192.168.9.0 [110/4] via 192.168.0.3, 00:39:26, FastEthernet1/0
    192.168.10.0/30 is subnetted, 1 subnets
O IA    192.168.10.0 [110/3] via 192.168.0.3, 00:39:26, FastEthernet1/0
    192.168.11.0/30 is subnetted, 1 subnets
O IA    192.168.11.0 [110/3] via 192.168.0.3, 00:39:26, FastEthernet1/0
    192.168.4.0/30 is subnetted, 1 subnets
R        192.168.4.0 [120/1] via 192.168.1.2, 00:00:06, FastEthernet0/0
    192.168.5.0/30 is subnetted, 1 subnets
O IA    192.168.5.0 [110/2] via 192.168.0.3, 00:39:28, FastEthernet1/0
    192.168.6.0/30 is subnetted, 1 subnets
O IA    192.168.6.0 [110/3] via 192.168.0.3, 00:39:28, FastEthernet1/0
    192.168.7.0/30 is subnetted, 1 subnets
O IA    192.168.7.0 [110/4] via 192.168.0.3, 00:39:29, FastEthernet1/0
```

```

C    192.168.0.0/24 is directly connected, FastEthernet1/0
    192.168.1.0/30 is subnetted, 1 subnets
C    192.168.1.0 is directly connected, FastEthernet0/0
    192.168.2.0/30 is subnetted, 1 subnets
R    192.168.2.0 [120/1] via 192.168.1.2, 00:00:07, FastEthernet0/0
    192.168.3.0/30 is subnetted, 1 subnets
R    192.168.3.0 [120/2] via 192.168.1.2, 00:00:07, FastEthernet0/0

R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

```

Gateway of last resort is not set

```

    192.168.8.0/30 is subnetted, 1 subnets
R    192.168.8.0 [120/1] via 192.168.1.1, 00:00:25, FastEthernet2/0
    192.168.9.0/30 is subnetted, 1 subnets
R    192.168.9.0 [120/1] via 192.168.1.1, 00:00:25, FastEthernet2/0
    192.168.10.0/30 is subnetted, 1 subnets
R    192.168.10.0 [120/1] via 192.168.1.1, 00:00:25, FastEthernet2/0
    192.168.11.0/30 is subnetted, 1 subnets
R    192.168.11.0 [120/1] via 192.168.1.1, 00:00:25, FastEthernet2/0
    192.168.4.0/30 is subnetted, 1 subnets
C    192.168.4.0 is directly connected, FastEthernet0/0
    192.168.5.0/30 is subnetted, 1 subnets
R    192.168.5.0 [120/1] via 192.168.1.1, 00:00:00, FastEthernet2/0
    192.168.6.0/30 is subnetted, 1 subnets
R    192.168.6.0 [120/1] via 192.168.1.1, 00:00:01, FastEthernet2/0
    192.168.7.0/30 is subnetted, 1 subnets
R    192.168.7.0 [120/1] via 192.168.1.1, 00:00:01, FastEthernet2/0
    192.168.1.0/30 is subnetted, 1 subnets
C    192.168.1.0 is directly connected, FastEthernet2/0
    192.168.2.0/30 is subnetted, 1 subnets
C    192.168.2.0 is directly connected, FastEthernet1/0
    192.168.3.0/30 is subnetted, 1 subnets
R    192.168.3.0 [120/1] via 192.168.2.2, 00:00:18, FastEthernet1/0

```

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

PC1:

PC1> ping 192.168.4.2

```

84 bytes from 192.168.4.2 icmp_seq=1 ttl=62 time=39.662 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=62 time=25.436 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=62 time=24.650 ms

```

```
84 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=25.728 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=62 time=25.265 ms
```

```
PC1> ping 192.168.6.2
```

```
84 bytes from 192.168.6.2 icmp_seq=1 ttl=59 time=61.393 ms
84 bytes from 192.168.6.2 icmp_seq=2 ttl=59 time=55.866 ms
84 bytes from 192.168.6.2 icmp_seq=3 ttl=59 time=55.888 ms
84 bytes from 192.168.6.2 icmp_seq=4 ttl=59 time=55.824 ms
84 bytes from 192.168.6.2 icmp_seq=5 ttl=59 time=55.517 ms
```

```
PC1> ping 192.168.9.2
```

```
84 bytes from 192.168.9.2 icmp_seq=1 ttl=58 time=80.308 ms
84 bytes from 192.168.9.2 icmp_seq=2 ttl=58 time=65.934 ms
84 bytes from 192.168.9.2 icmp_seq=3 ttl=58 time=65.478 ms
84 bytes from 192.168.9.2 icmp_seq=4 ttl=58 time=65.574 ms
84 bytes from 192.168.9.2 icmp_seq=5 ttl=58 time=66.150 ms
```

```
PC1> ping 192.168.11.2
```

```
84 bytes from 192.168.11.2 icmp_seq=1 ttl=59 time=79.358 ms
84 bytes from 192.168.11.2 icmp_seq=2 ttl=59 time=55.120 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=59 time=55.898 ms
84 bytes from 192.168.11.2 icmp_seq=4 ttl=59 time=55.832 ms
84 bytes from 192.168.11.2 icmp_seq=5 ttl=59 time=56.433 ms
```

```
PC2:
```

```
PC2> ping 192.168.3.2
```

```
84 bytes from 192.168.3.2 icmp_seq=1 ttl=62 time=29.945 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=62 time=25.273 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=62 time=24.445 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=62 time=25.266 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=62 time=25.616 ms
```

```
PC2> ping 192.168.9.2
```

```
84 bytes from 192.168.9.2 icmp_seq=1 ttl=59 time=58.444 ms
84 bytes from 192.168.9.2 icmp_seq=2 ttl=59 time=55.737 ms
84 bytes from 192.168.9.2 icmp_seq=3 ttl=59 time=56.044 ms
84 bytes from 192.168.9.2 icmp_seq=4 ttl=59 time=54.594 ms
84 bytes from 192.168.9.2 icmp_seq=5 ttl=59 time=66.137 ms
```

```
PC2> ping 192.168.11.2
```

```
84 bytes from 192.168.11.2 icmp_seq=1 ttl=60 time=50.182 ms
84 bytes from 192.168.11.2 icmp_seq=2 ttl=60 time=65.595 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=60 time=45.120 ms
84 bytes from 192.168.11.2 icmp_seq=4 ttl=60 time=45.698 ms
84 bytes from 192.168.11.2 icmp_seq=5 ttl=60 time=46.467 ms
```

PC2> ping 192.168.6.2

84 bytes from 192.168.6.2 icmp_seq=1 ttl=60 time=45.715 ms
84 bytes from 192.168.6.2 icmp_seq=2 ttl=60 time=46.136 ms
84 bytes from 192.168.6.2 icmp_seq=3 ttl=60 time=45.998 ms
84 bytes from 192.168.6.2 icmp_seq=4 ttl=60 time=45.538 ms
84 bytes from 192.168.6.2 icmp_seq=5 ttl=60 time=44.793 ms

PC3:

PC3> ping 192.168.3.2

84 bytes from 192.168.3.2 icmp_seq=1 ttl=58 time=79.732 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=58 time=65.779 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=58 time=75.444 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=58 time=65.397 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=58 time=75.523 ms

PC3> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=59 time=69.872 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=59 time=54.879 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=59 time=66.390 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=59 time=56.276 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=59 time=66.235 ms

PC3> ping 192.168.11.2

84 bytes from 192.168.11.2 icmp_seq=1 ttl=62 time=35.471 ms
84 bytes from 192.168.11.2 icmp_seq=2 ttl=62 time=25.633 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=62 time=25.222 ms
84 bytes from 192.168.11.2 icmp_seq=4 ttl=62 time=25.783 ms
84 bytes from 192.168.11.2 icmp_seq=5 ttl=62 time=25.178 ms

PC3> ping 192.168.6.2

84 bytes from 192.168.6.2 icmp_seq=1 ttl=60 time=52.234 ms
84 bytes from 192.168.6.2 icmp_seq=2 ttl=60 time=46.001 ms
84 bytes from 192.168.6.2 icmp_seq=3 ttl=60 time=45.719 ms
84 bytes from 192.168.6.2 icmp_seq=4 ttl=60 time=45.365 ms
84 bytes from 192.168.6.2 icmp_seq=5 ttl=60 time=45.425 ms

PC4:

PC4> ping 192.168.3.2

84 bytes from 192.168.3.2 icmp_seq=1 ttl=59 time=58.361 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=59 time=55.565 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=59 time=55.144 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=59 time=55.144 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=59 time=54.693 ms

PC4> ping 192.168.4.2

```
84 bytes from 192.168.4.2 icmp_seq=1 ttl=60 time=44.205 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=60 time=45.733 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=60 time=45.436 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=60 time=45.303 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=60 time=45.497 ms
```

```
PC4> ping 192.168.9.2
```

```
84 bytes from 192.168.9.2 icmp_seq=1 ttl=62 time=28.709 ms
84 bytes from 192.168.9.2 icmp_seq=2 ttl=62 time=24.687 ms
84 bytes from 192.168.9.2 icmp_seq=3 ttl=62 time=25.857 ms
84 bytes from 192.168.9.2 icmp_seq=4 ttl=62 time=25.537 ms
84 bytes from 192.168.9.2 icmp_seq=5 ttl=62 time=25.574 ms
```

```
PC4> ping 192.168.6.2
```

```
84 bytes from 192.168.6.2 icmp_seq=1 ttl=61 time=34.854 ms
84 bytes from 192.168.6.2 icmp_seq=2 ttl=61 time=35.940 ms
84 bytes from 192.168.6.2 icmp_seq=3 ttl=61 time=35.101 ms
84 bytes from 192.168.6.2 icmp_seq=4 ttl=61 time=35.338 ms
84 bytes from 192.168.6.2 icmp_seq=5 ttl=61 time=35.149 ms
```

```
PC5:
```

```
PC5> ping 192.168.9.2
```

```
84 bytes from 192.168.9.2 icmp_seq=1 ttl=60 time=49.473 ms
84 bytes from 192.168.9.2 icmp_seq=2 ttl=60 time=46.108 ms
84 bytes from 192.168.9.2 icmp_seq=3 ttl=60 time=45.587 ms
84 bytes from 192.168.9.2 icmp_seq=4 ttl=60 time=46.416 ms
84 bytes from 192.168.9.2 icmp_seq=5 ttl=60 time=46.171 ms
```

```
PC5> ping 192.168.3.2
```

```
84 bytes from 192.168.3.2 icmp_seq=1 ttl=59 time=51.873 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=59 time=55.820 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=59 time=66.352 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=59 time=65.735 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=59 time=55.345 ms
```

```
PC5> ping 192.168.4.2
```

```
84 bytes from 192.168.4.2 icmp_seq=1 ttl=60 time=43.844 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=60 time=56.073 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=60 time=45.639 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=60 time=45.719 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=60 time=45.467 ms
```

```
PC5> ping 192.168.9.2
```

```
84 bytes from 192.168.9.2 icmp_seq=1 ttl=60 time=52.097 ms
84 bytes from 192.168.9.2 icmp_seq=2 ttl=60 time=45.113 ms
```

```

84 bytes from 192.168.9.2 icmp_seq=3 ttl=60 time=45.491 ms
84 bytes from 192.168.9.2 icmp_seq=4 ttl=60 time=45.188 ms
84 bytes from 192.168.9.2 icmp_seq=5 ttl=60 time=45.870 ms

```

```
PC5> ping 192.168.11.2
```

```

84 bytes from 192.168.11.2 icmp_seq=1 ttl=61 time=41.571 ms
84 bytes from 192.168.11.2 icmp_seq=2 ttl=61 time=35.184 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=61 time=35.397 ms
84 bytes from 192.168.11.2 icmp_seq=4 ttl=61 time=35.588 ms
84 bytes from 192.168.11.2 icmp_seq=5 ttl=61 time=35.719 ms

```

```

PC1 - PuTTY
4 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=25.728 ms
4 bytes from 192.168.4.2 icmp_seq=5 ttl=62 time=25.265 ms
C1> ping 192.168.6.2
4 bytes from 192.168.6.2 icmp_seq=1 ttl=59 time=61.393 ms
4 bytes from 192.168.6.2 icmp_seq=2 ttl=59 time=55.866 ms
4 bytes from 192.168.6.2 icmp_seq=3 ttl=59 time=55.088 ms
4 bytes from 192.168.6.2 icmp_seq=4 ttl=59 time=55.824 ms
4 bytes from 192.168.6.2 icmp_seq=5 ttl=59 time=55.517 ms
C1> ping 192.168.9.2
4 bytes from 192.168.9.2 icmp_seq=1 ttl=58 time=80.308 ms
4 bytes from 192.168.9.2 icmp_seq=2 ttl=58 time=65.934 ms
4 bytes from 192.168.9.2 icmp_seq=3 ttl=58 time=65.478 ms
4 bytes from 192.168.9.2 icmp_seq=4 ttl=58 time=65.574 ms
4 bytes from 192.168.9.2 icmp_seq=5 ttl=58 time=66.150 ms
C1> ping 192.168.11.2
4 bytes from 192.168.11.2 icmp_seq=1 ttl=59 time=69.872 ms
4 bytes from 192.168.11.2 icmp_seq=2 ttl=59 time=54.879 ms
4 bytes from 192.168.11.2 icmp_seq=3 ttl=59 time=66.390 ms
4 bytes from 192.168.11.2 icmp_seq=4 ttl=59 time=56.276 ms
4 bytes from 192.168.11.2 icmp_seq=5 ttl=59 time=66.235 ms
C3> ping 192.168.11.2
4 bytes from 192.168.11.2 icmp_seq=1 ttl=62 time=35.471 ms
4 bytes from 192.168.11.2 icmp_seq=2 ttl=62 time=25.633 ms
4 bytes from 192.168.11.2 icmp_seq=3 ttl=62 time=25.222 ms
4 bytes from 192.168.11.2 icmp_seq=4 ttl=62 time=25.783 ms
4 bytes from 192.168.11.2 icmp_seq=5 ttl=62 time=25.178 ms
C3> ping 192.168.6.2
4 bytes from 192.168.6.2 icmp_seq=1 ttl=60 time=52.234 ms
4 bytes from 192.168.6.2 icmp_seq=2 ttl=60 time=46.001 ms
4 bytes from 192.168.6.2 icmp_seq=3 ttl=60 time=45.719 ms
4 bytes from 192.168.6.2 icmp_seq=4 ttl=60 time=45.365 ms
4 bytes from 192.168.6.2 icmp_seq=5 ttl=60 time=45.425 ms

PC2 - PuTTY
84 bytes from 192.168.11.2 icmp_seq=2 ttl=60 time=65.595 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=60 time=45.120 ms
84 bytes from 192.168.11.2 icmp_seq=4 ttl=60 time=45.698 ms
84 bytes from 192.168.11.2 icmp_seq=5 ttl=60 time=46.467 ms
PC2> ping 192.168.5.2
84 bytes from 192.168.5.2 icmp_seq=1 ttl=252 time=31.331 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=252 time=35.625 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=252 time=35.477 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=252 time=35.358 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=252 time=36.028 ms
PC2> ping 192.168.6.2
84 bytes from 192.168.6.2 icmp_seq=1 ttl=60 time=45.715 ms
84 bytes from 192.168.6.2 icmp_seq=2 ttl=60 time=46.136 ms
84 bytes from 192.168.6.2 icmp_seq=3 ttl=60 time=45.998 ms
84 bytes from 192.168.6.2 icmp_seq=4 ttl=60 time=45.538 ms
84 bytes from 192.168.6.2 icmp_seq=5 ttl=60 time=44.793 ms

PC3 - PuTTY
4 bytes from 192.168.4.2 icmp_seq=1 ttl=59 time=69.872 ms
4 bytes from 192.168.4.2 icmp_seq=2 ttl=59 time=54.879 ms
4 bytes from 192.168.4.2 icmp_seq=3 ttl=59 time=66.390 ms
4 bytes from 192.168.4.2 icmp_seq=4 ttl=59 time=56.276 ms
4 bytes from 192.168.4.2 icmp_seq=5 ttl=59 time=66.235 ms
C3> ping 192.168.11.2
4 bytes from 192.168.11.2 icmp_seq=1 ttl=62 time=35.471 ms
4 bytes from 192.168.11.2 icmp_seq=2 ttl=62 time=25.633 ms
4 bytes from 192.168.11.2 icmp_seq=3 ttl=62 time=25.222 ms
4 bytes from 192.168.11.2 icmp_seq=4 ttl=62 time=25.783 ms
4 bytes from 192.168.11.2 icmp_seq=5 ttl=62 time=25.178 ms
C3> ping 192.168.6.2
4 bytes from 192.168.6.2 icmp_seq=1 ttl=60 time=52.234 ms
4 bytes from 192.168.6.2 icmp_seq=2 ttl=60 time=46.001 ms
4 bytes from 192.168.6.2 icmp_seq=3 ttl=60 time=45.719 ms
4 bytes from 192.168.6.2 icmp_seq=4 ttl=60 time=45.365 ms
4 bytes from 192.168.6.2 icmp_seq=5 ttl=60 time=45.425 ms

PC4 - PuTTY
84 bytes from 192.168.4.2 icmp_seq=1 ttl=60 time=44.205 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=60 time=45.733 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=60 time=45.436 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=60 time=45.303 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=60 time=45.497 ms
PC4> ping 192.168.9.2
84 bytes from 192.168.9.2 icmp_seq=1 ttl=62 time=28.709 ms
84 bytes from 192.168.9.2 icmp_seq=2 ttl=62 time=24.687 ms
84 bytes from 192.168.9.2 icmp_seq=3 ttl=62 time=25.857 ms
84 bytes from 192.168.9.2 icmp_seq=4 ttl=62 time=25.537 ms
84 bytes from 192.168.9.2 icmp_seq=5 ttl=62 time=25.574 ms
PC4> ping 192.168.6.2
84 bytes from 192.168.6.2 icmp_seq=1 ttl=61 time=34.854 ms
84 bytes from 192.168.6.2 icmp_seq=2 ttl=61 time=35.940 ms
84 bytes from 192.168.6.2 icmp_seq=3 ttl=61 time=35.101 ms
84 bytes from 192.168.6.2 icmp_seq=4 ttl=61 time=35.338 ms
84 bytes from 192.168.6.2 icmp_seq=5 ttl=61 time=35.144 ms

PC5 - PuTTY
3 *192.168.5.1 19.481 ms (ICMP type:3, code:1, Dest
)
PC5> ping 192.168.9.2
84 bytes from 192.168.9.2 icmp_seq=1 ttl=60 time=49.473 ms
84 bytes from 192.168.9.2 icmp_seq=2 ttl=60 time=46.108 ms
84 bytes from 192.168.9.2 icmp_seq=3 ttl=60 time=45.587 ms
84 bytes from 192.168.9.2 icmp_seq=4 ttl=60 time=46.416 ms
84 bytes from 192.168.9.2 icmp_seq=5 ttl=60 time=46.171 ms
PC5> ping 192.168.3.2
84 bytes from 192.168.3.2 icmp_seq=1 ttl=59 time=51.873 ms
84 bytes from 192.168.3.2 icmp_seq=2 ttl=59 time=55.820 ms
84 bytes from 192.168.3.2 icmp_seq=3 ttl=59 time=66.352 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=59 time=65.735 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=59 time=55.345 ms
PC5> ping 192.168.4.2
84 bytes from 192.168.4.2 icmp_seq=1 ttl=60 time=43.844 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=60 time=56.073 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=60 time=45.639 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=60 time=45.719 ms
84 bytes from 192.168.4.2 icmp_seq=5 ttl=60 time=45.467 ms
PC5> ping 192.168.9.2
84 bytes from 192.168.9.2 icmp_seq=1 ttl=60 time=52.097 ms
84 bytes from 192.168.9.2 icmp_seq=2 ttl=60 time=45.113 ms
84 bytes from 192.168.9.2 icmp_seq=3 ttl=60 time=45.491 ms
84 bytes from 192.168.9.2 icmp_seq=4 ttl=60 time=45.188 ms
84 bytes from 192.168.9.2 icmp_seq=5 ttl=60 time=45.870 ms
PC5> ping 192.168.11.2
84 bytes from 192.168.11.2 icmp_seq=1 ttl=61 time=41.571 ms
84 bytes from 192.168.11.2 icmp_seq=2 ttl=61 time=35.184 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=61 time=35.397 ms
84 bytes from 192.168.11.2 icmp_seq=4 ttl=61 time=35.588 ms
84 bytes from 192.168.11.2 icmp_seq=5 ttl=61 time=35.719 ms

```

Рисунок 5.1 – Ура пинг везде идет!

6) Перехватить в wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

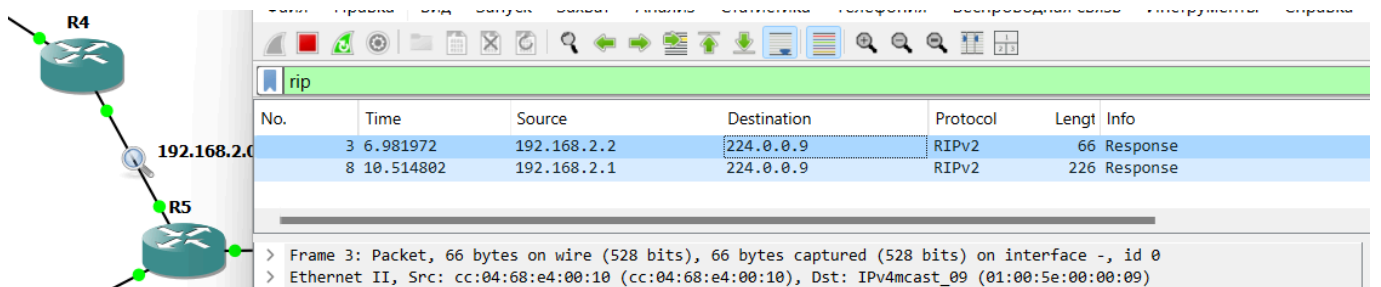


Рисунок 6.1 – Захват RIP сообщений.

```
Internet Protocol Version 4, Src: 192.168.2.2, Dst: 224.0.0.9
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    Total Length: 52
    Identification: 0x0000 (0)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 2
    [Expert Info (Note/Sequence): "Time To Live" != 1 for a packet sent to the Local Network Control Block (see RFC 3171)]
    ["Time To Live" != 1 for a packet sent to the Local Network Control Block (see RFC 3171)]
    [Severity level: Note]
    [Group: Sequence]
  Protocol: UDP (17)
  Header Checksum: 0x1546 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.2.2
  Destination Address: 224.0.0.9
  [Stream index: 0]
User Datagram Protocol, Src Port: 520, Dst Port: 520
  Source Port: 520
  Destination Port: 520
  Length: 32
  Checksum: 0x933f [unverified]
  [Checksum Status: Unverified]
  [Stream index: 0]
  [Stream Packet Number: 1]
  [Timestamps]
  > UDP payload (24 bytes)
```

Рисунок 6.2 – IP и UDP заголовки.

В IPv4 заголовке выделено, что пакет имеет TTL больше 1. Адрес отправителя – маршрутизатор R4, адрес получателя – специальный многоадресный (224.0.0.9) зарезервированный для RIP. Для UDP протокола используются специальные порт 520, опять же зарезервированные для RIP.

На рисунке 6.3 представлено сообщение от R4, который сообщает о том что ему известна сеть 192.168.3.0. Это сеть в которой находится PC1. Уточняется версия протокола, адрес, маска, метрика – 1 хоп.

```
Routing Information Protocol
  Command: Response (2)
  Version: RIPv2 (2)
  > IP Address: 192.168.3.0, Metric: 1
    Address Family: IP (2)
    Route Tag: 0
    IP Address: 192.168.3.0
    Netmask: 255.255.255.252
    Next Hop: 0.0.0.0
    Metric: 1
```

Рисунок 6.3 – Сообщение от R4.

Но R5 вещает о всех сетях которые ему известны, поэтому его сообщение RIP больше (рисунок 6.4). Тут можно увидеть, что для адресов 192.168.1.0 и 192.168.4.0 метрика указана 1. В сообщении отправленном маршрутизатором R5 адресов больше, так как он связывает R4 с остальной сетью. Для удаленных более чем на один маршрутизатор сетей метрика указана больше.

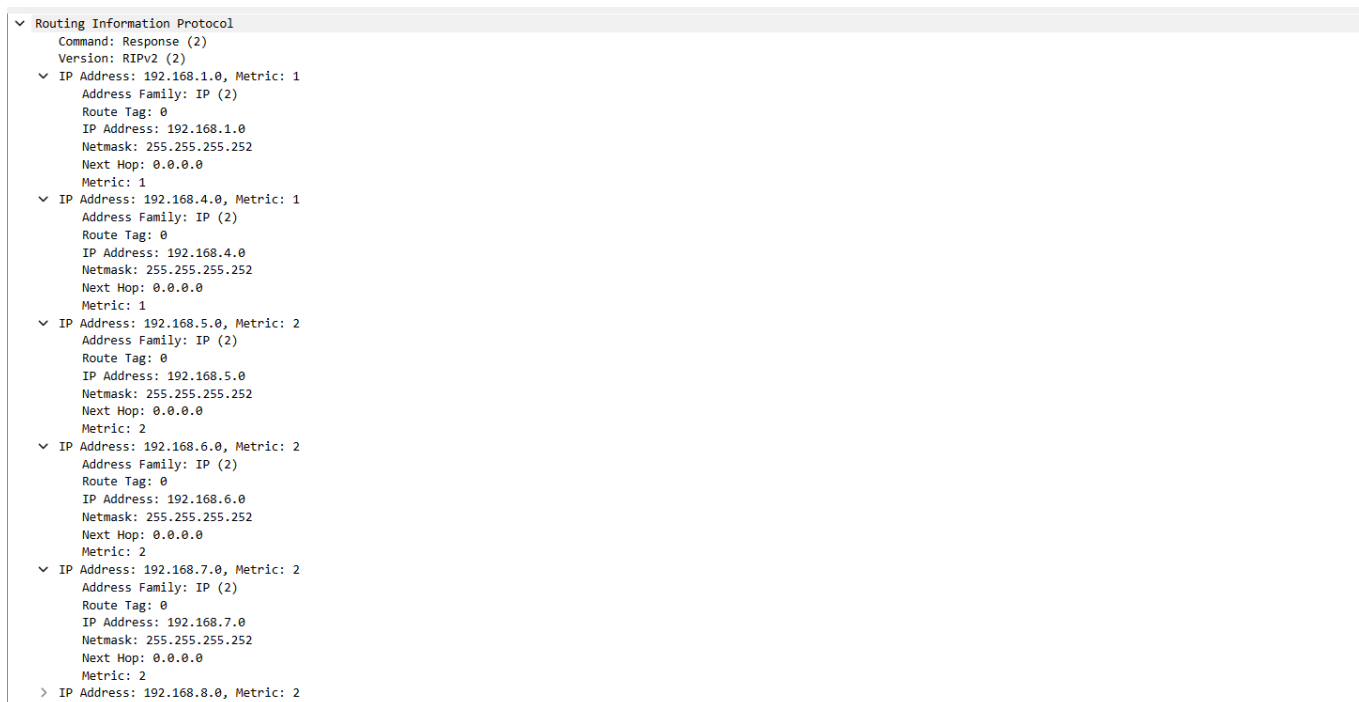


Рисунок 6.4 – RIP сообщение от R5.

Теперь рассмотрим пакеты протокола OSPF (рисунок 6.5).

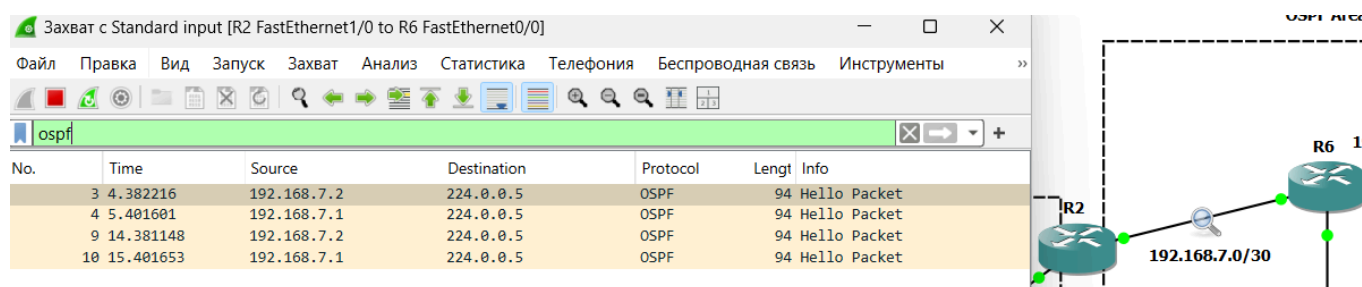


Рисунок 6.5 – Захват OSPF пакетов.

На рисунке 6.6 приведен IP заголовок Hello Packet от 192.168.7.2 (маршрутизатор R6). Адрес получателя указан стандартный выделенный для OSPF протокола. При этом тут не используется UDP или TCP, тип протокола указан OSPF IGP.

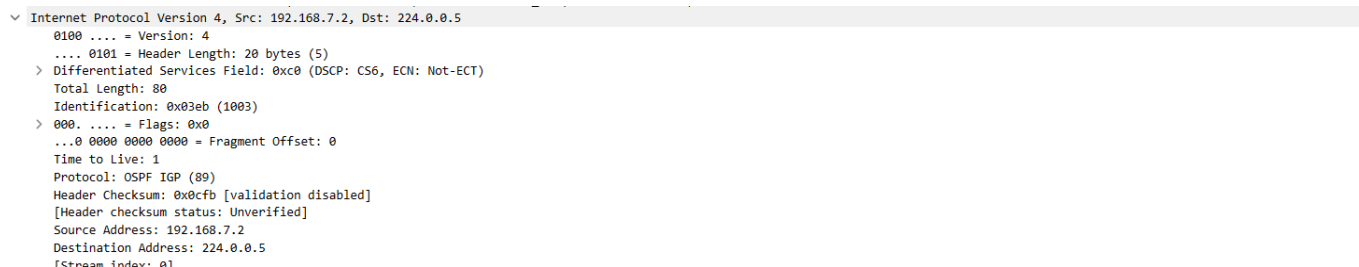


Рисунок 6.6 – IP заголовок OSPF Hello Packet.

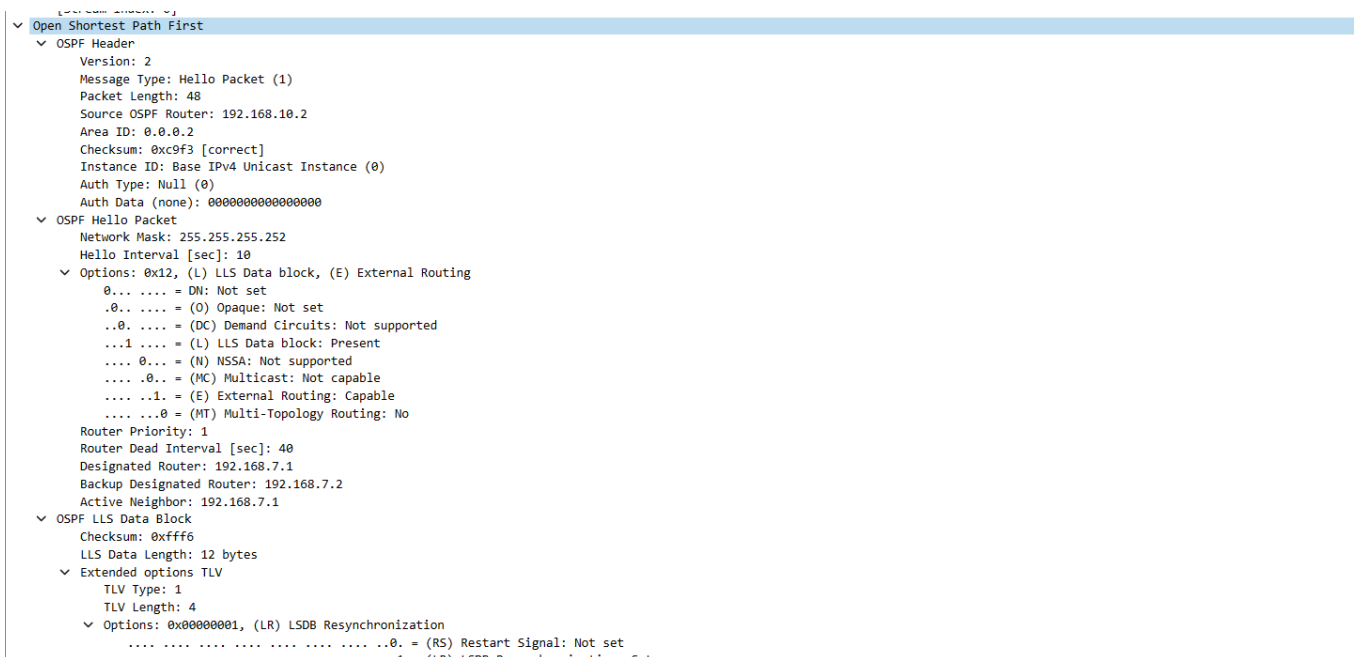


Рисунок 6.7 – Заголовок OSPF.

Заголовок OSPF разделен на несколько частей: OSPF Header, OSPF Hello Packet, OSPF LLS Data Block.

Header содержит:

- Версия протокола: 2
- Тип сообщения (hello)
- Длина пакета
- Адрес отправителя
- ID зоны, (зона 2)

Hello Packet содержит:

- Маску подсети,
- Интервал hello сообщений
- Опции
- Интервал времени, через который можно считать маршрутизатор недостижимым
- Главный маршрутизатор, он же помечен как активный сосед.